End-to-End Experimental Workflow: AST-Driven Multi-Paradigm OLAP Benchmarking (SF=1, 10, 100 GB)

PHASE 1: Multi-Scale TPC-H Provisioning (SF = 1, 10, 100 GB)

- Multi-Scale Datasets: SF=1 (8.66M rows), SF=10 (86.6M), SF=100 (866M rows)
- Local Columnar: ZSTD Parquet + DuckDB Native + Apache Arrow Streaming
- Local Row-Store Baseline: SQLite 3 B-Tree with PK/FK Indexes
- Cloud Serverless: Google BigQuery Unoptimized vs. Clustered (US Multi-Region)
- External Validation Suite: Official 22-Query TPC-H Benchmark (Q1–Q22)

PHASE 2: 75-Query Factorial Workload Design (25 Seeds × 3 Variations)

- 5 Structural Tiers (5 Canonical Seeds/Tier = 25 Independent Seeds):
  L1 (Scan/Filter), L2 (4–8 Way Joins), L3 (Staged CTEs), L4 (Window), L5 (UNNEST)
- 3 Selectivity Perturbations per Seed (Total N = 75 Factorial Queries):
  V1: High Selectivity (1–5%) | V2: Medium (15–30%) | V3: Full-Horizon (65–95%)
- Orthogonal Design: Decouples AST operator depth from predicate selectivity

PHASE 3: Static Abstract Syntax Tree (AST) Profiling & Revised CSCI_v2 Quantification (SQLGlot Parser)

- Parses all 75 factorial queries + 22 official TPC-H queries into canonical AST representations to extract 7 operator dimensions:
  Tree Depth D(q), Total Operator Nodes N(q), Join Count J(q), Nested CTE/Subquery Depth S(q), Window Specs W(q), Array UNNEST U(q), Branches C(q)
- Revised Composite Structural Complexity Index: CSCI_v2(q) = 1.5 D(q) + 1.0 log2 N(q) + 2.5 J(q) + 3.0 S(q) + 3.5 W(q) + 4.0 U(q) + 1.5 C(q)
- Tier Progression: Mean CSCI_v2 progresses from 27.9 (L1) and 36.2 (L2) to 41.4 (L3), 41.7 (L4), and 43.6 (L5) [Validated via ±50% Monte Carlo Sensitivity]

Paradigm 1: Row-Store

- Engine: SQLite 3 (ANSI C)
- Storage: B-Tree Row-Store
- Execution: Tuple-at-a-Time
  Volcano Iterator Model
- Baseline Mean: 5071.1 ms
  (Throughput: 0.064 GB/s)

Paradigm 2: Serverless Cloud

- Engine: Google BigQuery
- Storage: Capacitor Columnar
  (Unopt: 2190.6ms, Clust: 745.4ms)
- Controls: Cache Disabled,
  Zero-Billing Sandbox Quota
- Pruning: 78.4% in V1 ($1.47/1M)

Paradigm 3: Vectorized CPU

- 1) DuckDB (C++ Columnar)
  Parquet (254.4ms) vs Native (183.7ms)
  SF100 Out-of-Core Spill: 43.8 GB
- 2) Polars SQL (Rust Arrow: 66.5ms)
- 3) Apache DataFusion (Rust: 88.2ms)
  Peak L5 UNNEST: 18.87 GB/s

Paradigm 4: SIMT CUDA GPU

- Engine: PyTorch CUDA Tensor
- GPU: NVIDIA RTX 2050 (4GB VRAM)
- 3 Fairness Modes (M1, M2, M3)
- Mean PCIe H2D: 34.71 ms
- Mean Warm Kernel: 22.25 ms
- SF10/100 Batched PCIe Stream

PHASE 5: Hierarchical Statistical Verification & Fairness

- 6-Engine Row-Level Multiset Equivalence Audit: EX = 100.0% (tol <= 1e-6)
- Hierarchical Inference: 25-Seed Bootstrap 95% CIs vs. 75-Query Wilcoxon
- Two-Way Non-Parametric Interaction: Scheirer-Ray-Hare H & ART F-Tests
- Three-Mode CPU/GPU Fairness: Kernel-only (M1), Transfer (M2), End-to-End (M3)

PHASE 6: AST-Aware Query Router & Cryptographic Audit

- Learned AST Cost Router (5-Fold Out-of-Seed CV): 45.09 ms Mean Latency
  (4.07x Speedup vs. DuckDB, 74.7% Exact Match, 81.3% Within-5ms Optimal)
- 19-File SHA-256 Telemetry Audit Certificate (REPRODUCTION_CERTIFICATE_V2.md)
- Open-Source Benchmark Repository & Journal of Big Data Manuscript:
  https://github.com/preecha-msu/bigquery-olap-benchmark